6. Affine Cipher — Breaking the Code

CODE:

```
from math import gcd

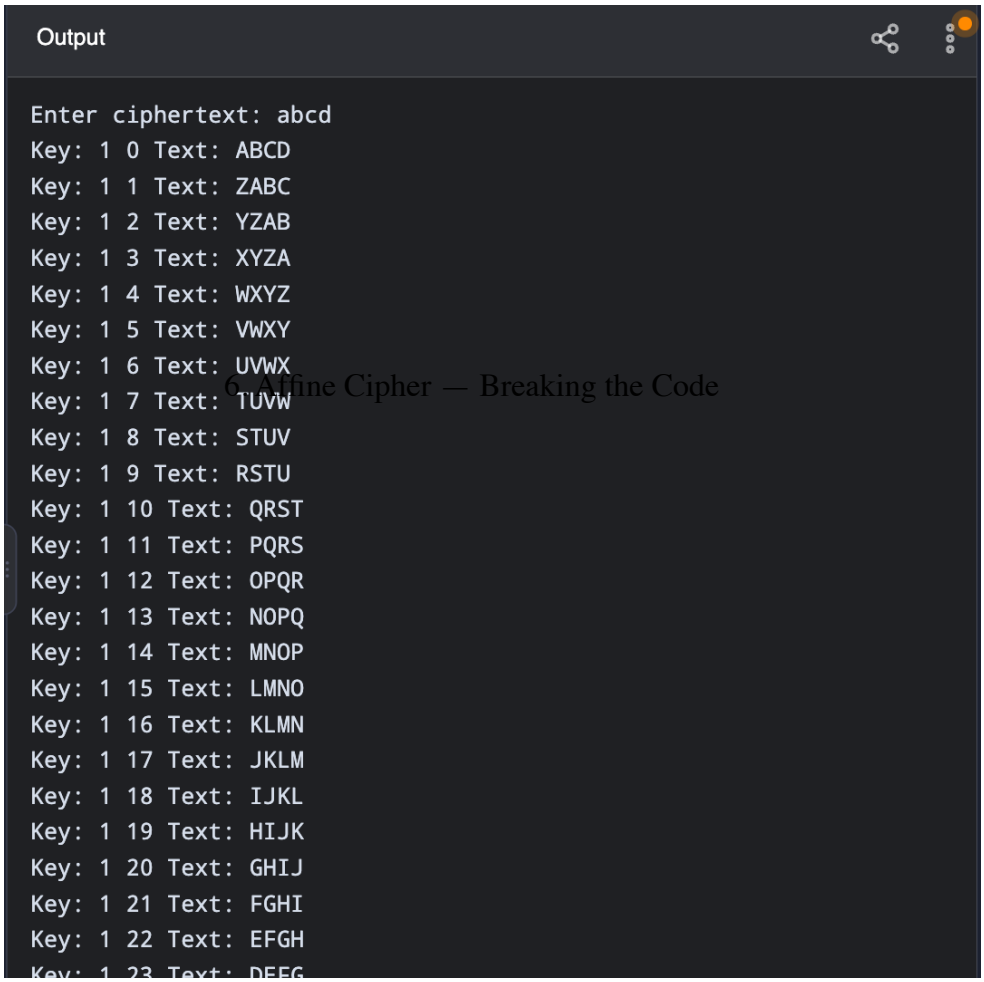
ciphertext = input("Enter ciphertext: ").upper()

for a in range(1, 26):
    if gcd(a, 26) == 1:
        for b in range(26):
            plaintext = ""

            for ch in ciphertext:
                if ch.isalpha():
                    c = ord(ch) - ord('A')
                    inverse = pow(a, -1, 26)
                    p = (inverse * (c - b)) % 26
                    plaintext += chr(p + ord('A'))
                else:
                    plaintext += ch

            print("Key:", a, b, "Text:", plaintext)
```

OUTPUT:



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